Julien Bernardi

GitHub Portfolio
LinkedIn

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Montmorency, France

Education

- **Université PSL**[1] — Paris, France
  MSc (Year 2) — Artificial Intelligence, Systems and Data · ENS | Mines Paris | Paris-Dauphine — *Sept 2026 – Sept 2027*
  - **Main courses**: Machine Learning, Deep Learning, Generative AI, Reinforcement Learning, Natural Language Processing, Advanced Databases, Optimization for Machine Learning.
- **Université Paris-Dauphine PSL** — Paris, France
  MSc (Year 1) — Computer Science, Decision and Data — *Aug 2025 – July 2026*
  - **Main courses**: Algorithmic and Advanced Programming, Mathematics for Data Science, Collective and Multicriteria Decision, Combinatorial Optimization, Database Management Systems, Machine Learning.
- **Université Paris-Dauphine PSL** — Paris, France
  Double Bachelor's Degree in Artificial Intelligence and Economics — *Sept 2022 – June 2025*
  - **Main courses**: Deep Learning, Statistical Learning, Machine Learning, Statistical Modeling, Algorithms and Data Structures, Linear algebra, Econometrics, Game Theory, Probability Theory, Optimization

[1] PSL is ranked #34 worldwide (QS World University Rankings 2027).

Projects (cf. GitHub)

- **Abdominal Aortic Aneurysm HISA Score Prediction**: Predicting aortic wall inflammation from clinical data and histological image embeddings (April 2026)
- **Lymph Node Metastasis Detection – ENS Challenge Data**: Automated detection of lymph node metastases in breast cancer patients using different Machine Learning models (April – May 2025)
- **Enhancing GANs with Discriminator Rejection Sampling**: Implemented Discriminator Rejection Sampling (DRS) and adaptive truncation to improve GANs, enhancing sample quality and diversity in images. (Nov – Dec 2024)
- **Fraud Detection using Machine Learning**: Developed a fraud detection model using Gradient Boosting and Random Forest, achieving 1st place in the ENS Challenge Data competition (June 2024)

Experience

- **AI Engineer (Apprenticeship)** — Palaiseau, France
  Ministerial Agency for Defense AI (AMIAD) – French Ministry of the Armed Forces — *Sept 2026 – Sept 2027*
  - Member of the artificial intelligence research division
- **AI Engineer Intern – Trustworthy Neural Text Classification** — Montrouge, France
  Institut National de la Statistique et des Études Économiques (INSEE) – SSP Lab — *April 2026 – July 2026*
  - Designed and compared explainable text classification architectures to automatically assign an activity code (Nomenclature d'Activités Françaises) to newly created companies
- **Research Intern – Predictive Approaches for Financial Time Series** — Tours, France
  Université de Tours, Laboratoire d'Informatique Fondamentale et Appliquée de Tours (LIFAT) — *June 2025 – July 2025*
  - Studied financial time series and developed predictive models to analyze market trends and enhance understanding of complex financial data

Skills

- **Programming Languages**: Python, C, SQL, R
  - **Machine Learning**: PyTorch, PyTorch Lightning, scikit-learn, numpy, huggingface, pandas, Neural Networks (CNNs, Encoders/Decoders, GANs), Natural Language Processing (RNN, LSTM, Transformers)
  - **MLOps & Experimentation**: MLflow, Docker, Git/GitHub, Conda, Linux/Shell Scripting, uv
- **Data Science**: Statistics, Hypothesis Testing, Bayesian Methods, Time-Series, OOP, Linear Programming, Excel, Jupyter Notebooks
- **Languages**: French (native), English (fluent, C1)

Volunteering

- **France Bénévolat** — Paris, France
  Assistant Marketing — *Sept – Oct 2020*
  - **Graphic Design**: Created visual materials and graphic elements for various projects
- **ALIAS** — Paris, France
  Communication Specialist — *Oct 2024 – July 2025*
  - **Visual Identity**: Designed visuals to support ALIAS initiatives and strengthen the association's communication